



OPEN Blockchain framework with IoT device using federated learning for sustainable healthcare systems

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The Internet of Medical Things (IoMT) sector has advanced rapidly in recent years, and security and privacy are essential considerations in the IoMT due to the extensive scope and implementation of IoMT networks. Machine learning (ML) and blockchain (BC) technologies have dramatically improved the functionalities and services of Healthcare 5.0, giving rise to a new domain termed Smart Healthcare. A proactive healthcare system may prevent long-term harm by recognizing issues early. This would improve patients' quality of life while alleviating their worry and healthcare expenses. The IoMT facilitates several capabilities in information technology, including intelligent and interactive healthcare. Consolidating medical information into a singular repository to train a robust ML model engenders apprehensions around privacy, ownership, and adherence to regulatory standards due to increased concentration. Federated learning (FL) addresses previous challenges using a centralized aggregate server to distribute global learning models. The local participant controls patient data, ensuring data confidentiality and security. Hence, this study proposes the Federated Blockchain-IoT Framework for Sustainable Healthcare Systems (FBCI-SHS) for a secure health monitoring system. Additionally, this paper presents the Intrusion Detection System (IDS) as a tool for healthcare network intrusion detection, allowing doctors to track patients' vitals using medical sensors and anticipate when they could become sick so they can take preventative steps. The suggested system proves that the method is well-suited for medical monitoring. In contrast, the high prediction accuracy for intrusion detection and the high efficiency in disease detection achieved by the proposed FBCI-SHS healthcare 5.0 system. The proposed method achieves data privacy and security by 98.73%, intrusion detection efficiency by 97.16%, disease detection accuracy by 96.425, proactive healthcare management by 98.37%, and interoperability by 96.74%.

Keywords Blockchain, Internet of things (IoT), Federated learning, Sustainable healthcare systems, Intrusion detection system, Smart healthcare, Monitoring system

Smart Healthcare systems, an important part of Healthcare 5.0, have evolved as a direct outcome of healthcare transformation through the fast development of the IoMT¹. Such systems combine smart, interactive technology to provide efficient, proactive, and individualized care². The IoMT would greatly advance healthcare quality by linking sensors, medical devices, and healthcare networks and enabling real-time monitoring, diagnosis, and treatment³. Data security, privacy, and regulatory compliance are three main areas where such an enormous scale poses significant difficulties related to IoMT networks⁴. All these challenges needed to be handled properly to adequately use IoMT for the long-term sustainability of health care systems⁵. ML has immensely improved healthcare services by detecting diseases, predicting results, and optimizing treatment plans by analyzing massive health data⁶.

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Consolidated sensitive medical data into central repositories, problems with ownership, patient privacy, and compliance with regulations such as GDPR and HIPAA arise⁷. Federated Learning offers an interesting alternative where local nodes like IoT devices and hospitals can collaborate to train a global ML model without exchanging raw data⁸. In this way, model performance is high, and data confidentiality and security are guaranteed due to its decentralized mode of operation⁹. Blockchain technology can be a very good addition to FL concerning the trust and integrity of data in IoMT networks¹⁰. For example, blockchain will be an excellent choice for storing sensitive medical information because it offers decentralized ledger technology¹¹. Transparent access management and secure, tamper-proof data storage¹². The best features of blockchain and blockchain technology are combined to create a solid framework that overcomes the shortcomings in traditional healthcare systems¹³.

This paper proposes the FBCI-SHS¹⁴ to increase healthcare efficiency, privacy, and security. FL, blockchain, and IoMT are framework components¹⁵. Due to the framework using medical sensors from Internet of Things devices for real-time health monitoring, doctors can track patients' vital signs and predict ailments. Early identification and therapy using FBCI-SHS may improve patient outcomes, save healthcare costs, and reduce clinician and patient anxiety¹⁶. This suggests that the proposed intrusion detection system (IDS) protects the health network against cyberattacks¹⁷. The suggested FBCI-SHS system works well in Healthcare 5.0 to forecast incursions and diagnose disorders¹⁸.

Motivation

The urgency of improving the safety, confidentiality, and functioning of medical technology-based healthcare systems prompted this research. The architecture includes blockchain technology and federated learning to improve patient experience and build intelligent, ever-growing Healthcare 5.0 systems. The framework allows safe health monitoring and medical data handling by resolving confidentiality, trust, and compliance standards.

Problem statement

Concerning data security, privacy preservation, and regulatory compliance, traditional healthcare systems based on the IoMT have considerable constraints. Due to their reliance on centralized designs, these systems are very vulnerable to data breaches, unwanted access, and single points of failure. These security holes reduce healthcare networks' responsiveness, scalability, and efficiency while jeopardizing patient privacy. Distributed IoMT devices provide an increasing amount of heterogeneous medical data, which the centralized method cannot handle. Because of this, clinical decision-making suffers from decreased predictive accuracy, slower diagnoses, and restricted interoperability. A strong, decentralized solution that protects user privacy is urgently needed to address these problems. To tackle these issues, this study presents a new way to combine blockchain technology with federated learning and the IoT. The goal is to improve the system's scalability, trust, and operational efficiency while establishing a safe, transparent, and highly predictive framework for healthcare monitoring. This framework must also guarantee compliance with privacy legislation.

Contributions

The Contribution of this paper of this paper is given in Table 1.

The revolutionary possibilities of combining FL and blockchain with IoT for long-term healthcare sustainability to provide a safe and intelligent answer to the current problems in Smart Healthcare systems. A component of the proposed system plays a crucial role in protecting the health network from cyber-attacks. Through secure communication, the IDS enhances the reliability of the framework as it identifies potential attacks and minimizes them.

The rest of the paper is prearranged as follows: [Related works](#) deliberates the related works, [Proposed method](#) suggests the FBCI-SHS model, [Result and discussion](#) reflects the experimental outcomes, and [Conclusion](#) concludes the research article.

Related works
ML in healthcare system (ML-HCS)

New and amazing treatment methods have come due to the fast development of technology. These methods intend to increase scalability and security while boosting patient care. This particular abstract describes a healthcare system transformation system utilizing blockchain technology combined with hybrid deep learning (DL) techniques. According to Ali A. et al.¹⁹, blockchain technology provides a transparent and decentralized infrastructure to facilitate secure data storage, sharing, and access management and reduces Data Privacy and Security by 33.53%. Combining DL and ML algorithms enables hybrid DL to process complex healthcare data such as medical images, records, and sensor data quickly and efficiently. The newly developed permissions-based blockchain architecture and a hybrid DL model provide modern healthcare systems with the requisite security and scalability. The model provides strong privacy protection for patients by giving access only to authenticated

Development of a novel framework	Combining federated learning, blockchain, and IoMT to enhance healthcare systems' security, privacy, and efficiency, ensuring decentralized and secure medical data management.
Introduction of an IDS to safeguard healthcare networks	Detecting and mitigating cyber threats and ensuring reliable communication between IoT devices, FL servers, and blockchain networks.
Demonstration of the framework's effectiveness	Through high prediction accuracy in intrusion detection and disease diagnosis, enabling proactive health monitoring and early interventions for improved patient outcomes.

Table 1. Contribution of this paper.

users while enabling improved collaboration and information sharing among healthcare providers Arza, M. S. et al.²⁰.

Hybrid DL methods (HDLM)

Smartified the healthcare system by incorporating IoT and ubiquitous computing into medical devices. More than just healing patients is the modern purpose of healthcare. An SHS system uses patients' implanted medical devices and wearables to identify and prevent serious diseases in real-time, reducing intrusion detection efficiency by 32.86%. In several ways, bad actors may take advantage of a SHS, and the security risks are growing in tandem with its capabilities by Sundas, A. et al.,²¹. Among them include, but are not limited to, tampering with medical equipment, injecting false data to alter vital signs, and disrupting normal SHS functioning. Using ML to detect potentially dangerous user activities, this work introduces HealthGuard, a novel security framework for SHSs. To distinguish between normal and unhealthy activities, HealthGuard analyzes the vital signs of several devices linked to SHS by Kumar, R. et al.²².

Big data analysis (BDA)

Transmission or sharing of health data and data outside of secure big data threatens patient privacy. This is because they start to hide their phases and become more marginalized due to infractions. These kinds of phases will have detrimental effects on scientific research. This problem is addressed by proposing a Secure Blockchain System for the Management and Sharing of Electronic Medical Records in the Big Data Subject by Marichamy, V. S. et al.²³. There is a separation of sensitive and non-sensitive information in the big data set obtained from the healthcare facility. The blockchain system verifies the authenticity of transactions using asymmetric cryptography. In this case, the user key is generated using a secured bitwise cryptographic hash generator and may be used to access the newly added record. A user must submit a request to CHG on the blockchain system whenever they want to access data from a healthcare application by Pablo, R. G. J. et al.,²⁴.

Artificial neural network (ANN)

Patients may be notified of particular health issues or used as a tool for therapy or follow-up with wearable healthcare technology. The security of these gadgets is a rising problem due to the proliferation of technology and connection. The use of healthcare systems provided by the IoT is greatly jeopardized due to inexperienced users' lack of security knowledge and the possibility of several intermediate attacks that might compromise health information by Kumar M. et al.²⁵. Investigates innovative healthcare solutions that provide scalable, efficient, and secure systems by combining blockchain technology with hybrid DL. Combining blockchain security and hybrid DL improves data processing while ensuring user privacy. In addition to enhancing patient care and system security, AI models such as HealthGuard mitigate security threats in smart healthcare systems and reduce accurate disease detection by 39.05%. Ali Akbar Movassagh et al.²⁶ suggested the Artificial neural networks training algorithm integrating invasive weed optimization with a differential evolutionary model. This study uses meta-heuristics to train a perceptron neural network more accurately. An integrated technique was used to calculate neural network input coefficients. The recommended strategy was compared to invasive weed optimization and ant colonies to evaluate its performance. According to the data, the recommended strategy converges faster with the neural network coefficient than others. However, the recommended method decreased neural network prediction error.

Nazir, Sajid et al.²⁷ proposed the Blockchain of things for healthcare asset management. Critical asset data is vulnerable to eavesdropping and alteration either in transit or at rest, which poses privacy and security issues when data is gathered and transferred by radio, Bluetooth, or other communication technologies. For the sake of privacy and tamper-proofing, the asset information is crucial. Combining blockchain technology with the IoT may achieve high trust and security. Thus, only authorized parties may safely access the data inside the private blockchain.

Jafar A. Alzubi²⁸ discussed the Blockchain-based Lamport Merkle Digital Signature in IoT healthcare. To begin authenticating IoT devices, the Lamport Merkle Digital Signature Generation (LMDSG) model builds a tree with the leaves representing the hash function of sensitive patient medical data. More specifically, a CHC employs Lamport Merkle Digital Signature Verification (LMDSV) to ascertain the origin of the LMDSG.

Mehdi Gheisari et al.²⁹ introduced the ontology-based privacy-preserving (OBPP) framework in IoT-based smart city. Addressing heterogeneity while protecting the privacy of IoT devices is the goal of the first module's ontology, which is a data storage model. Second, the author has rules for semantic reasoning that we may use to address service quality and identify unusual patterns. The third component is a privacy rules manager that can adapt to the ever-changing privacy behaviors of IoT devices, making it easier to deal with the difficulties of protecting user privacy. Comprehensive simulations on a fictitious smart city dataset show that our method outperforms the existing alternatives while being affordable and resistant to data leaks. Its broad applicability to smart cities is a result.

The summary of related work is given in Table 2.

Proposed method

The proposed FBCI-SHS addresses critical privacy, security, and efficiency challenges in IoMT-based smart healthcare. Integrating federated learning, blockchain, and IoT ensures secure health monitoring, intrusion detection, and proactive patient care. Utilizing blockchain's consensus methods and immutable ledger, the system greatly benefits from a decentralized environment, guaranteeing trust and transparency across federated IoT healthcare nodes. Blockchain technology makes decentralized identity management and safe aggregation of model changes possible, as opposed to public key infrastructure (PKI), which relies on centralized certificate authorities and is susceptible to problems like certificate revocation or single points of failure. In the FBCI-SHS

S. No.	Methods	Advantages	Limitations
1	ML in Healthcare System (ML-HCS) ¹⁹	Ensures secure, transparent, and decentralized data sharing Protects patient privacy Facilitates access management for authorized parties	Requires significant computational power Scalability challenges with large datasets Complex integration with existing healthcare systems
2	Hybrid DL Methods (HDLML) ²¹	Enables fast and accurate processing of complex healthcare data (e.g., medical records, images) Combines DL with traditional ML methods	Requires large amounts of labeled data Complex model training High computational resources needed for real-time analysis
3	Big Data Analysis (BDA) ²³	Protects patient privacy and ensures the authenticity of transactions Separates sensitive and non-sensitive data Improves research security	Performance may degrade with increased data size Complexity in managing access controls and cryptographic methods Regulatory compliance challenges
4	Artificial Neural Network (ANN) ²⁵	Allows for predictive health notifications and therapeutic follow-ups Enhances security for wearable healthcare devices	Vulnerable to attacks if not properly secured Requires expertise to train and implement ANN models effectively Potentially high energy consumption for real-time analysis

Table 2. The summary of related work.

architecture, blockchain is used for more than just authentication and encryption; it also keeps a verifiable and immutable record of access logs, model update hashes, and federated learning transactions. In addition, PKI systems aren't built to automate policy enforcement and anomaly reactions; smart contracts can. Therefore, although PKI might provide the groundwork for secure communication, blockchain's distributed trust and programmability qualities make it more suited to dynamic, large-scale healthcare settings with several stakeholders.

Secure and scalable system integration

The proposed FBCI-SHS architecture uses blockchain and federated learning to solve IoMT privacy, security, and regulatory issues. This solution decentralizes patient data, promotes medical data governance compliance, and prevents data confidentiality breaches.

Figure 1 shows a federated learning, blockchain, FBCI-SHS, architectural framework based on IoMT technologies. The framework has many layers addressing different but equally relevant safe and efficient healthcare delivery issues. The IoT Layer connects several medical sensors and devices to gather real-time patient data. This data is processed at the Federated Learning Aggregation Layer for local processing while protecting user privacy and avoiding storage security problems associated with centralized databases. A global model makes predictive precision and preventative health monitoring possible. It boosts credibility by protecting transactions with decentralized, immutable storage and guaranteeing data integrity.

This increased privacy protection and security management makes protecting sensitive patient data and complying with laws easier. Health data analytics may save lives by managing healthcare, detecting illnesses, and preventing network infiltration using pooled information. To guarantee authenticity and traceability, the FBCI-SHS framework uses cryptographic keys unique to each IoT device to sign model updates once trained locally. The signed changes are sent to the blockchain network, where smart contracts check the authenticity and validity of each model by comparing the hashes given with anticipated forms and confirming digital signatures. A change must pass this validation procedure to be posted to the blockchain ledger. A predetermined consensus technique, such as Practical Byzantine Fault Tolerance (PBFT), certifies the data's validity and allows for safe aggregation after enough validated changes have been logged. Additionally, the blockchain maintains the immutable audit trail of all modifications, which helps traceability and prevents tampering. This layered approach ensures smart healthcare systems' security, interoperability, and economy.

$$\partial \propto' \rightarrow Ns [ap' - sm] + 2vd [sw - qxa't'] - klo [jsr''] . \quad (1)$$

The variable vd encompasses factors $\partial \propto'$ that have an influence $Ns [ap' - sm]$ on illness detection accuracy and proactive healthcare management. Equation 1 ∂ indicates the confidentiality of data ($2vd [sw - qxa't']$), Ns reflects computer safety aspects (intrusion detecting efficiency of $klo [jsr'']$).

$$t_rEd \rightarrow nX [s - 8bw''] + Bas [si - slw''] - vs [a - nwqa''] . \quad (2)$$

The Eq. 2 is a representation of the historical reliability t_rEd and efficiency ($X [s - 8bw'']$) of the FBCI-SHS infrastructure in the context of managing healthcare operations. The variable $Bas [si - slw'']$ is used to represent network execution metrics, the variable $vs [a - nwqa'']$ is used to address blockchain-assisted security.

$$n_frd \rightarrow Ma [sz - 8vw''] - Cs [ew - am''] + Ksj'' [xwq''] . \quad (3)$$

In the Eq. 3, the variable n_frd represents the accuracy of ML in predicting illnesses, $Ma [sz - 8vw'']$ represents the cost savings that are gained $Ksj'' [xwq'']$ efficient processing of information. By guaranteeing that the framework can deliver healthcare solutions that are accurate, cost-effective, and secure, this equation contains the framework's capacity.

$$\forall as [ju - am''] \rightarrow kn [s - new''] + 8Vs [w - 9vwq''] . \quad (4)$$

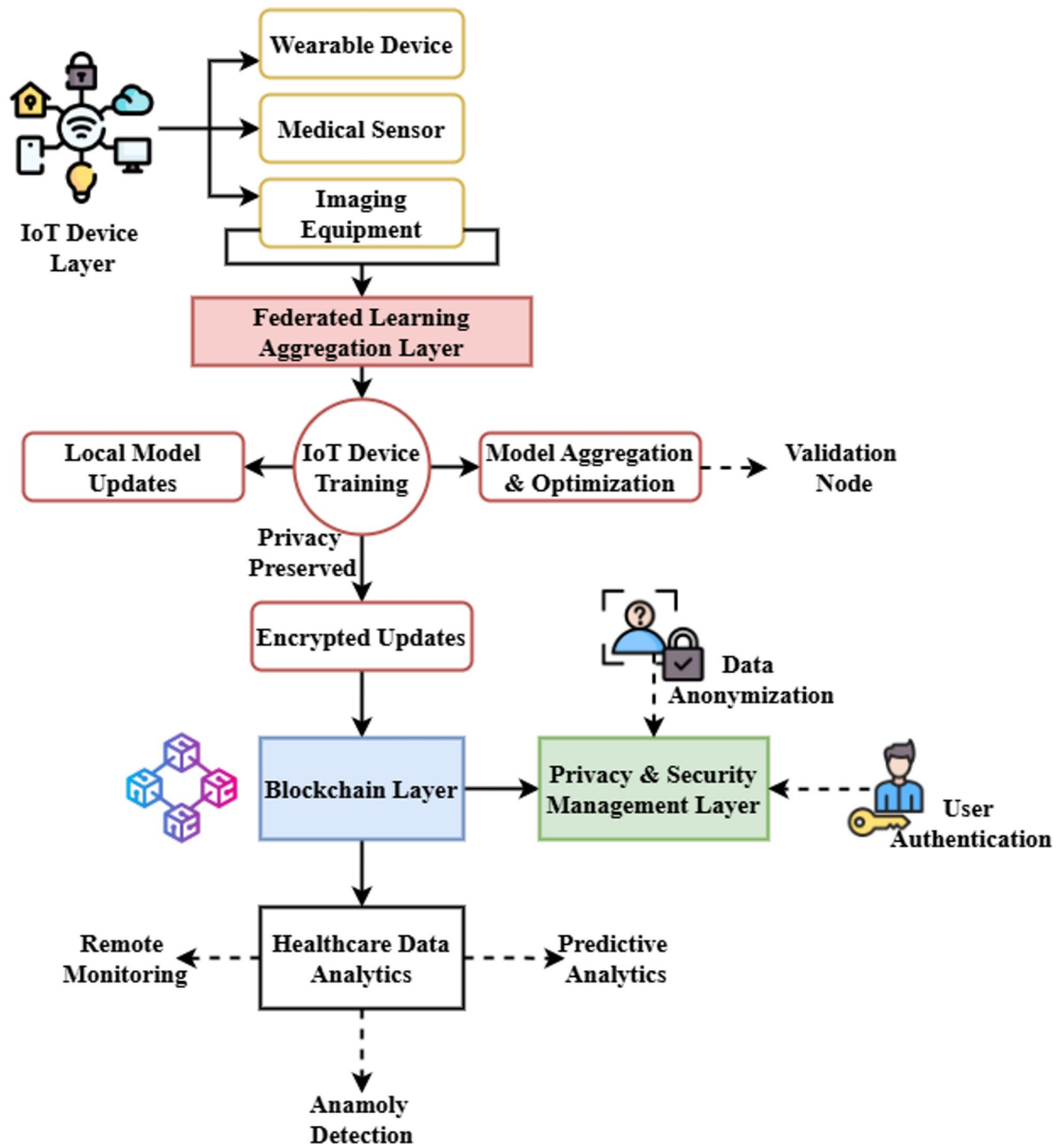


Fig. 1. Architecture of sustainable healthcare.

The Eq. 4 contains universal confidentiality of information ($\forall ds$) within the context of the FBCI-SHS. The variable $[ju - am']$ reflects the sharing of knowledge $8Vs[w - 9vwq']$ that is made possible by federated learning, and the variable $kn[s - new']$ measures the system's capacity to scale. This equation demonstrates the framework's capacity to safely handle and analyze data scattered across several locations.

$$u_v rD : \rightarrow nS [Av - am'] + 9B [Sx - wmq'] - csw' . \quad (5)$$

The FBCI-SHS framework takes into account both the user value ($nS [Av - am']$) and the distribution of resources $u_v rD$). Network scalability for medical device accommodation is denoted by $9B [Sx - wmq']$, security upgrades enabled by blockchain are represented by csw' , and computational efficiency. This equation highlights the framework's emphasis on healthcare benefits by guaranteeing safe, scalable, and resource-efficient

procedures. The inclusion of differential privacy was based on the assumption that all participating IoT devices and edge nodes acted honestly but curiously, following the federated learning protocol and maybe trying to deduce information from shared model changes. The communication network was deemed stable since no scenarios involving catastrophic delay, packet loss, or node failure were simulated.

It was assumed that all devices had synced clocks to ensure that federated rounds and blockchain consensus could be executed on time. Similarly, the simulation did not consider node turnover or dynamic reconfiguration; it assumed a constant number of devices (say, 100 nodes). Even though the data distribution among edge nodes was not uniform, it was nevertheless believed to be balanced enough to prevent training bias. There was zero tolerance for node-to-node coordination to poison models or damage the blockchain (e.g., 51% of attacks).

Patient monitoring methods of the FBCI-SHS framework are shown in Fig. 2a. Medical sensors linked to the IoT let physicians monitor patients from their homes. Blockchain technology ensures integrity and privacy, ensuring the data is delivered safely. Healthcare experts access the encrypted data via certified channels to make snap choices. Federated learning lets the system handle data in real-time without violating privacy. Thus, proactive healthcare management may be achieved without sacrificing data security and regulatory compliance.

$$dev := II' [Xsq'' + 9ve''] - Vs [se - sn''] + 9Vsw'' . \quad (6)$$

In the FBCI-SHS framework, the data efficacy (dev) and value ($II' [Xsq'' + 9ve'']$) are represented by the equation. Data security and processing speed are measured by $Vs[se - sn'']$ and $9Vsw''$, localized intelligence is represented. Effective and safe healthcare data management is ensured by this framework's capability to balance effectiveness, protection, and scalability, as shown by Eq. 6.

$$drvb := li [sf - 9bw''] + Ksa [ds - uyw''] - nsqa'' . \quad (7)$$

In the FBCI-SHS architecture, Eq. 7 represents the efficiency of the blockchain ($Ksa[ds - uyw'']$) and the dependability of the data ($drvb$). In this context, $li[sf - 9bw'']$ stands for secure federated educational systems at the local level, $nsqa''$ for confidentiality. The capability of the framework to provide efficient, safe, and dependable processing and monitoring of healthcare data is shown by this equation.

Figure 2b shows the Shard technique inside FBCI-SHS architecture: Data governance and access control are managed by the Authority Shard. This is maintained by government agencies such as the Ministry of Health. Healthcare organizations will manage the Cache Shard. Blockchain-based communication between shards will ensure secure and efficient data processing. This structure improves data integrity and confidentiality and prevents unauthorized access, a major concern in managing sensitive healthcare data.

$$nvd := Ns [w - 9vw''] + iw [q] - jw [kiy - aq] + jeq'' . \quad (8)$$

In the FBCI-SHS framework, the Eq. 8 stands for data efficiency ($Ns[w - 9vw'']$) and network value (nvd). Network safety and flexibility are represented by $iw[q]$, intelligent workflow integration is denoted by $jw[kiy - aq]$, computational load optimization is addressed by jeq'' , and equitable resource allocation. Equation highlights the framework's capacity to optimize network functioning, improve workflow intelligence, and process data.

$$m^f r [v - sm''] := ls [w - nqw''] + nd [rwaq'' - ndq] . \quad (9)$$

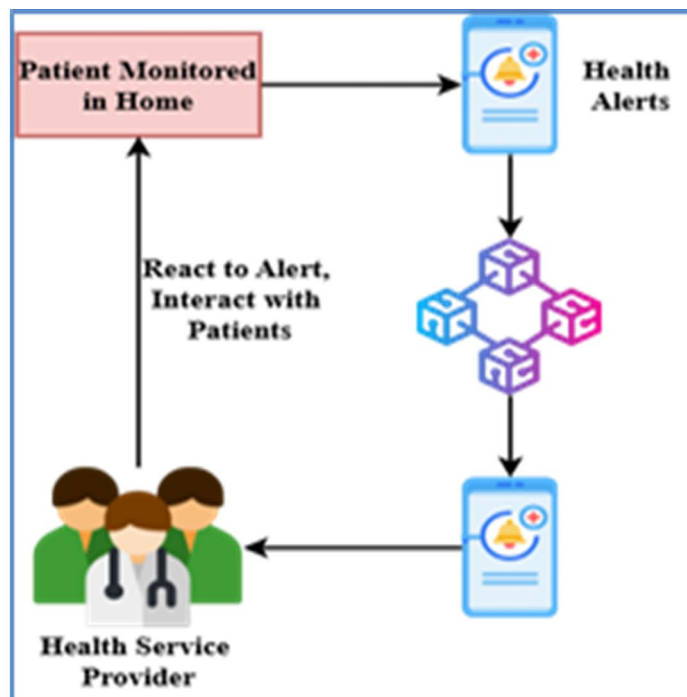
Inside the FBCI-SHS framework, Eq. 9 stands for the machine intelligence capabilities ($m^f r$) and dependability ($[v - sm'']$). Network fluctuations in distributing resources and anomaly detection are captured by $ls[w - nqw'']$ and localized system resilience and safety are denoted by $nd[rwaq'' - ndq]$. This equation shows that the framework uses ML to make healthcare systems more reliable.

Advanced intrusion and disease detection

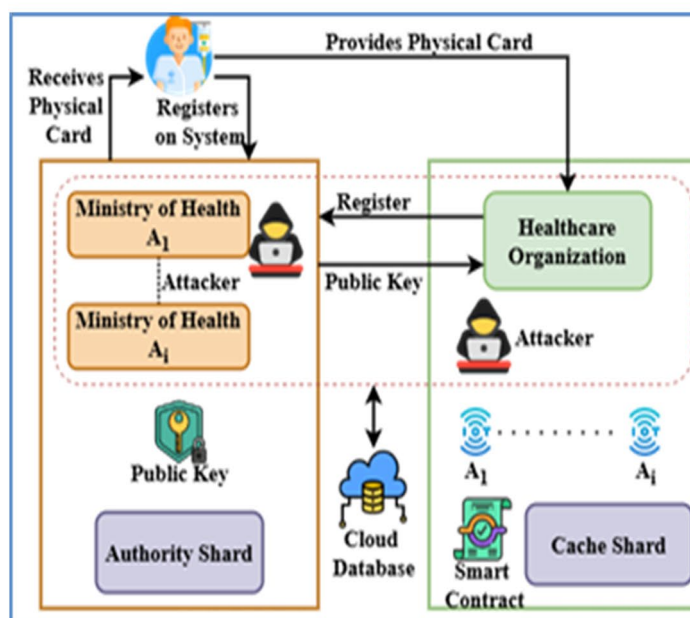
Having an Intrusion Detection System (IDS) in place allows for a 97.16% success rate in accurately identifying network threats. At the same time, the system detects diseases with a 96.42% accuracy rate, allowing clinicians to foresee potential dangers to their patient's health and suggest treatments promptly.

Figure 3 shows the Schematic of the proposed FBCI-SHS Federated Blockchain-IoT Framework for Sustainable Healthcare Systems. The Health Edge is a central server that coordinates federated learning operations and ensures secure aggregation and model updates. Data collection from IoT devices controlled by patients is done in a decentralized manner. In this sense, patients' health may be continuously watched in real-time without violating their rights. Patients, IoT devices, and healthcare providers center the network; blockchain technology guarantees unchangeable information and secure transactions. Every node in the global learning model does not broadcast personally identifiable patient information while involved. The design prioritizes privacy, encourages ownership, and ensures compliance with regulations. Health Edge equips medical professionals with proactive management tools that aid in detecting diseases and incursions. By working together, blockchain and federated learning build a smart healthcare ecosystem that is effective, secure, and robust; this, in turn, allows for high-performance prediction and proactive management.

$$jof [s - bwq''] := jds [6gew''] + od [s - nsq''] - snw'' . \quad (10)$$



(a)



(b)

Fig. 2. (a) Secure health monitoring and data sharing. (b) Decentralized sharing for secure data management.

The FBCI-SHS framework emphasizes the importance of scalability $d[s - nsq'']$ and joint data security snw'' , while operational information management $[s - bwq'']$ and communication synchronization are represented by jof and system-wide workload optimization is taken into consideration by $jds[6gew'']$. Equation 10 demonstrates resources, handles data securely, and keeps processes running smoothly.

$$p_d[ui - sn''] * g[snw'' + sj'] : \rightarrow ls[xwq - anf''] + Js[s - 8bw''] . \quad (11)$$

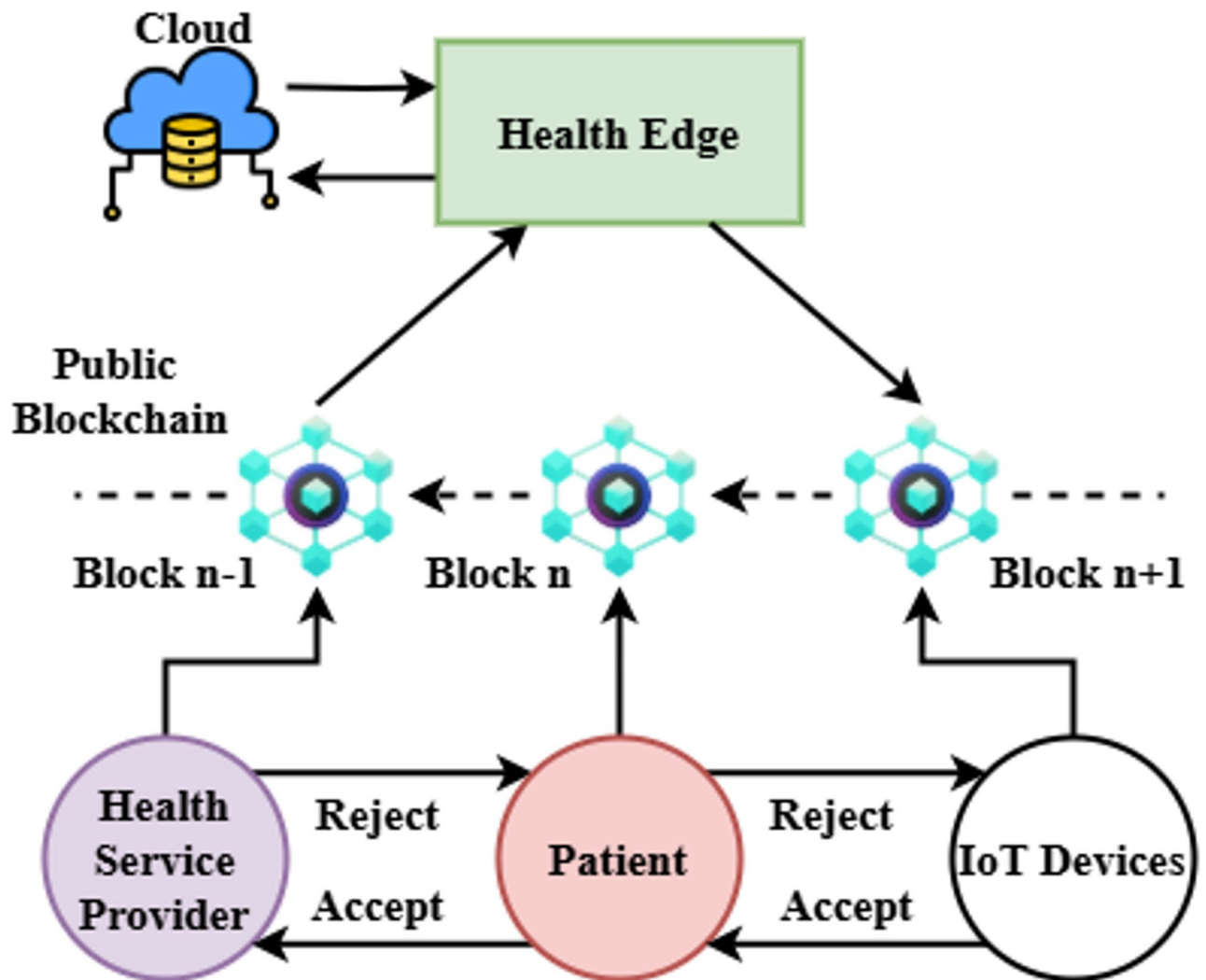


Fig. 3. Federated Blockchain-IoT integration for healthcare.

Within the FBCI-SHS framework, Eq. 11 (p_d) describes the prediction dynamics and data interface ($[ui - sn']$). $g[nsu' + sj']$ guarantees safe, scalable blockchain access, $Js[s - 8bv']$ concentrates on localized system optimization, and $lS[xwq - anf']$ symbolizes the integration of the entire world and joint data processing. This equation shows the predictive capabilities of the framework for proactive care administration.

$$k_{fe} [kui - an'] : \rightarrow Js[s - bnr'] + Bs[jw - naq'] . \quad (12)$$

Equation 12 in the FBCI-SHS paradigm specifies the important features k_{fe} and the efficiency ($[kui - an']$). The sentence means that $Js[s - bnr']$ symbolizes the secure handling of data and integration of systems via blockchain, and $Bs[jw - naq']$ is a representation of resources. This equation highlights the need to improve functional reliability, safety, and scalability for proactive healthcare solutions to succeed.

$$k_{fe} [bsle - ma'] : \rightarrow msr[s - 7vf'] + Js[4v - asm'] . \quad (13)$$

The equation outlines the essential features ($msr[s - 7vf']$) and performance k_{fe} of the FBCI-SHS system. $[bsle - ma']$ stands for the dependability and efficiency of the system when subjected to different healthcare loads, $Js[4v - asm']$ for the safe transfer of data and the effective distribution of resources. Functionality, assurances, and scalability are all carefully balanced in this equation, guaranteeing safe and effective healthcare administration.

Figure 4 shows the design of a Fog Cloud Agent that allows for effective and safe healthcare monitoring using the IoT. The Fog Cloud Agent is a middleman between medical applications and decentralized IoT modules (iTM 1, iTM 2, iTM 3), ensuring real-time replies, privacy, and smooth data processing. Various iTMs that focus on distinct features of health care are available; they may relate to e-hospital services and can even provide

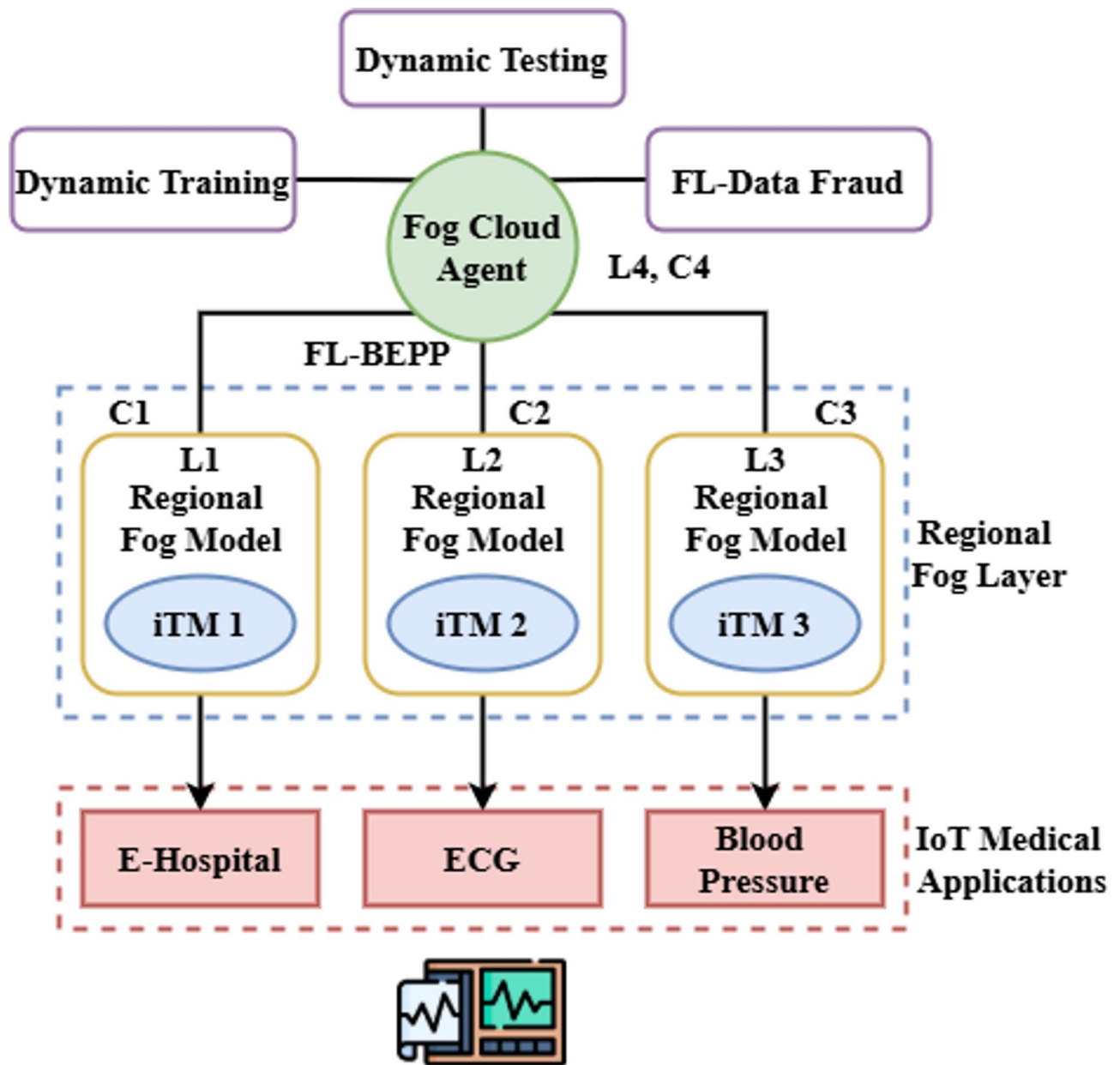


Fig. 4. Fog-based IoT integration for decentralized healthcare systems.

vital indicators such as ECG and blood pressure. So, the patient's data isn't directly fed into a centralized server for preserving their privacy. The model here utilizes federated learning, whereby it is trained across the entire world. It is known to reduce the healthcare application response times and minimize latency by compiling and interpreting local insights from iTMs. The process enables data security and integrity to be improved due to the adoption of blockchain technology. This healthcare ecosystem, designed with a scalable and decentralized architecture, achieves high efficiency and privacy requirements and stands out in proactive management and detection of illness and intrusions.

$$\partial_{ji} [\alpha' - vr'] := ns [r - vsw'] + jD [R - 7xsQ'] - ufvt. \quad (14)$$

The FBCI-SHS framework's incremental improvement (∂_{ji}) and intelligence ($[\alpha' - vr']$) are represented by the Eq. 14. Network flexibility $ns [r - vsw']$ and safety are represented by $jD [R - 7xsQ']$ and robust healthcare management is captured by $ufvt$. Highlighting the framework's ongoing development for safe and efficient healthcare monitoring deals with optimizing systems and computational efficiency.

$$i_f [si - beq'] * := mS [ju - sm'] + bs [w - 8va']. \quad (15)$$

The FBCI-SHS framework's interaction ability (i_f) and data consistency ($mS[ju - sm']$) are represented by the equation. $[si - beq']$ signifies scalability and system adaption driven by ML, whereas $bs[w - 8va']$ emphasizes safe access to the blockchain and data protection.

$$rv[f - bw'] : \rightarrow ndr[s - 8vw'] + fr[s - 8vwq'] - nsw'. \quad (16)$$

The FBCI-SHS framework's capabilities (rv) and reliability factor $f - bw'$ are represented by the above Eq. 16. Network data availability nsw' and safety are denoted by $ndr[s - 8vw']$ and fault resilience and adaptability in system performance are denoted by $fr[s - 8vwq']$.

$$k_{et} \ll nr - sn' \gg : \rightarrow Ns[nju - an'] + bsew[d - bqa']. \quad (17)$$

Key efficiency (k_{et} and reliability in networks $nr - sn'$ are both represented in the FBCI-SHS architecture by equality. $Ns[nju - an']$ represents the scalability and security of the network, while $bsew[d - bqa']$ represents safe information exchange.

Evaluation of proactive healthcare impact

The system's proactive healthcare management is made possible using IoMT sensors, as shown by the 98.37% increase in early intervention capabilities. By improving patient quality of life and guaranteeing sustainable and interoperable healthcare services, the assessment proves that FBCI-SHS is effective.

Figure 5 shows the Federated Blockchain-IoT Protecting patient privacy and enabling safe data gathering is the Sustainable Healthcare System (FBCI-SGS) goal, which includes medical IoT devices, patient sensors, and wearables in this ecosystem. Internet of Things (IoT) devices and wearables provide real-time data on vital signs to the localized edge components. Its edge components comprise a local Federated Learning (FL) model, storage, and an Intrusion Detection System (IDS). Thanks to cross-device collaboration and learning, this architecture

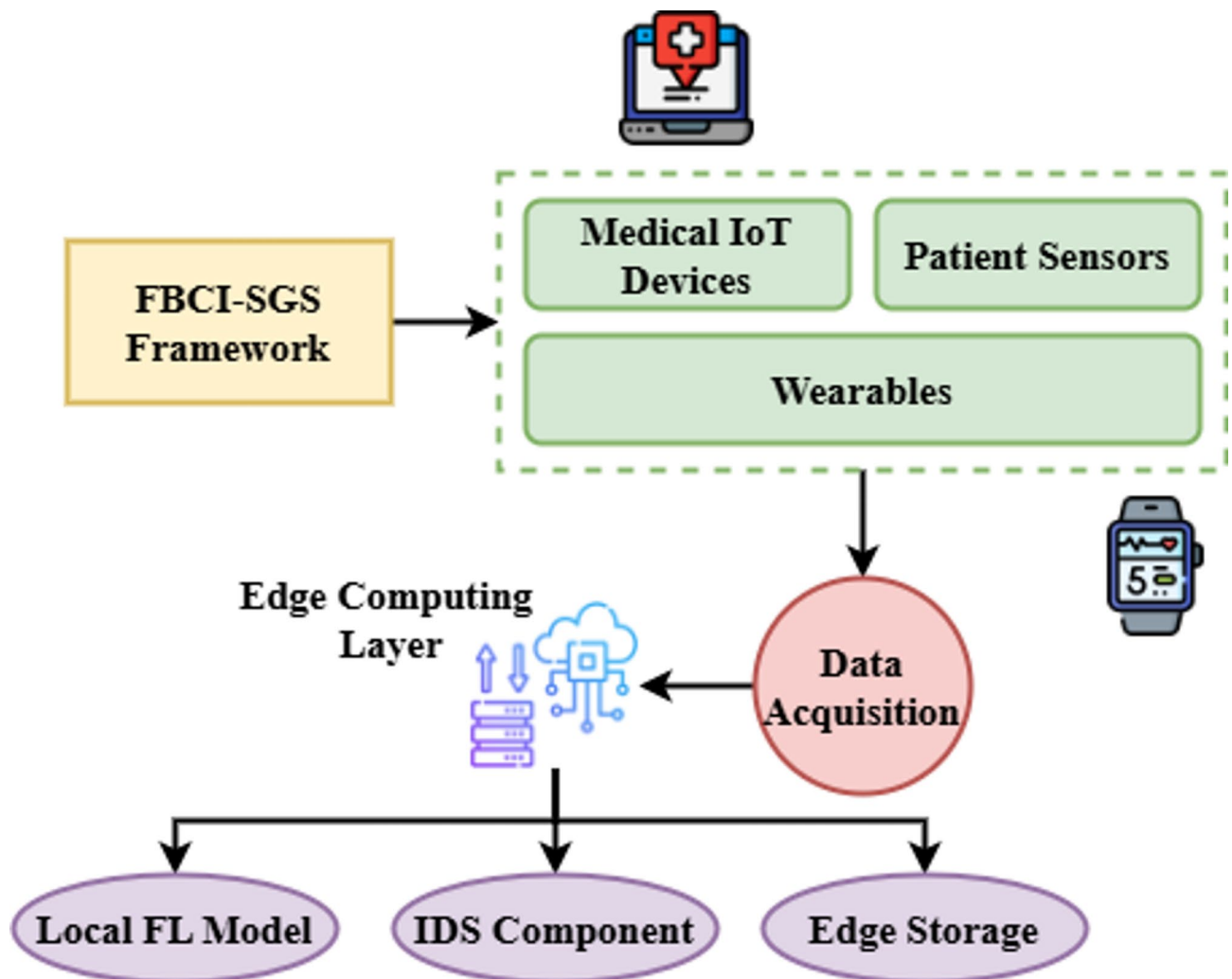


Fig. 5. Proposed method of federated Blockchain-IoT framework for sustainable healthcare systems.

in local FL keeps data secure and decentralized. Edge storage offers short-term, secure storage of collected data, while intrusion detection systems improve security by identifying any network breach. This design facilitates real-time decision-making by giving doctors useful patient health data. Health care service is further improved by the ability to take preventative measures. The system's connectivity with the Internet and federated learning allows it to achieve great efficiency, privacy, and security.

$$\tau_{\sigma} \rho \vartheta [\pi \tau \alpha' - dns'] \rightarrow bS [w - 8vq'] + ndr [s - bnq'] . \quad (18)$$

This Eq. 18 represents the FBCI-SHS framework's dynamic assessment ($bS [w - 8vq']$) and reliability of data ($\tau_{\sigma} \rho \vartheta$). $[\pi \tau \alpha' - dns'] \rightarrow$ stands for the blockchain's safekeeping and effective data transfer, while $ndr [s - bnq']$ is concerned with the dependability and safety of network data.

$$\partial_{ns}; [njs' - nr] \rightarrow Ns [we - 8br'] + nsw [nq - am'] . \quad (19)$$

In the FBCI-SHS framework, the Eq. 19 stands for the small improvement ($\partial_{ns}; [njs' - nr]$) in network reliability and security ($Ns [we - 8br']$).

$$\tau_{\rho} \tau \ll nj - gew' \gg \rightarrow ns [z - veq'] + jd [f - bsq'] . \quad (20)$$

In the FBCI-SHS framework, Eq. 20 represents both the changing efficiency ($nj - gew' \gg \rightarrow$) and the system's security ($\tau_{\rho} \tau$). Joint processing of information and securing the blockchain relies on storage are the main areas of attention for $ns [z - veq']$, while network reliability and efficacy are represented by $jd [f - bsq']$. Figure 6 shows the schematic diagram mapping equations to framework components.

Algorithm 1 shows the pseudocode of the Federated Blockchain-IoT Framework for Sustainable Healthcare Systems (FBCI-SHS).

Result and discussion

Dataset description

As explored in this dataset, sustainable healthcare systems may be created by integrating Blockchain technology with IoT devices utilizing Federated Learning. It includes blockchain-protected, anonymized patient data gathered from IoT health monitoring equipment. The data's emphasis on preserving data integrity and trust can improve health outcome prediction models, decrease energy usage, and protect patients' privacy³⁰. An IoT-based intensive care unit (ICU) with two beds and nine sensors per bed was the basis for our use case. The ICU also included a control unit, the Bedx-Control-Unit. These devices were all made with the help of the IoT-Flock tool. Table 3 gives the Simulation Environment. The performance of the proposed FBCI-SHS model has been analyzed based on metrics such as Data Privacy and Security, Intrusion Detection Efficiency, Disease Detection Accuracy, Proactive Healthcare Management, and interpretability compared to other models ML-HCS¹⁹, ANN²⁵, BDA²³ and HDLM²¹. This research applied a 5-fold stratified cross-validation protocol to maintain balanced class distributions across folds and simulate real-world federated learning environments by assigning data from different device types to separate clients. Evaluation metrics included classification accuracy, precision, recall, F1-score, AUC, and domain-specific metrics. This research defined the Early Anomaly Detection Rate (EADR) to quantify proactive healthcare management, calculated as the proportion of correctly identified anomalies within the first 10% of their appearance in the testing timeframe. Additionally, sustainability and scalability aspects were evaluated through communication overhead (in MB per federated round) and blockchain validation latency (in milliseconds), reflecting the real-time operational efficiency of FBCI-SHS in secure and responsive healthcare environments.

The data privacy and security analysis in the proposed FBCI-SHS clearly shows robust protection for patient information in Eq. 21. The method ensures 98.73% data privacy and security through federated learning, blockchain, and encryption. These technologies ensure that sensitive medical data is protected yet shared confidentially for healthcare monitoring and predictive analytics, as shown in Fig. 7.

$$k_{dr} [ju - sn'] \rightarrow nJ [fv - sn'] + U [s - 7bw'] - bsw' . \quad (21)$$

The data reliability (k_{dr}) and administration of resources ($[ju - sn']$) in the FBCI-SHS paradigm bsw' are represented by the Eq. 21. The network's combined security and functionality is represented by $nJ [fv - sn']$, and user-centric secure access is denoted by $U [s - 7bw']$. This equation shows how the trustworthiness of data, security, and effective use of resources are its top priorities in data privacy and security analysis.

According to the data concerning the performance of the FBCI-SHS, which concerns the analysis of intrusion detection, a high-performance index is gained with a detection rate of 97.16%. Advanced ML models and continuous supervision are implemented to detect unwanted intrusion or possible threats in Eq. 22. This helps maintain secure healthcare networks, making it possible to take proactive measures to increase the total system integrity and safety, as shown in Fig. 8.

$$Y_{fp} [b - snm'] \rightarrow ngf [\partial \alpha' - fsn'] + Js [s - 7bhw'] . \quad (22)$$

The data flow (Y_{fp}) and the performance of the system ($[b - snm']$) in the FBCI-SHS paradigm are symbolized by Eq. 22. To ensure the effective processing of data $[\partial \alpha' - fsn']$ and privacy protection $[\partial \alpha' - fsn']$, the system's flexibility and security are modeled by ngf , inside the blockchain network. Optimizing healthcare

Federated Blockchain-IoT Framework for Sustainable Healthcare Systems (FBCI-SHS)

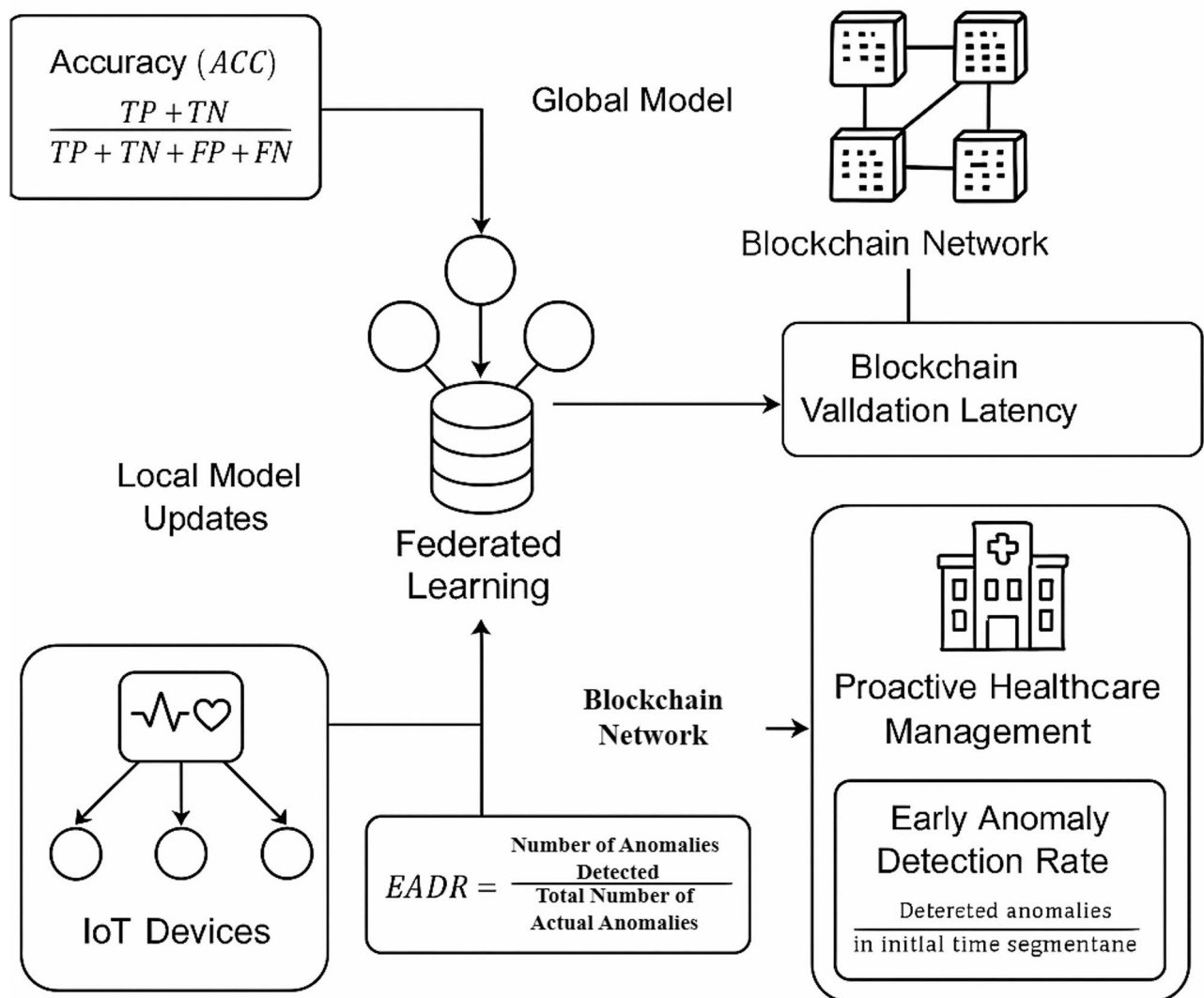


Fig. 6. Schematic diagram mapping equations to framework components.

monitoring while guaranteeing data security, scalability, and performance are highlighted by this equation in the analysis of intrusion detection efficiency.

The accuracy of disease detection in the FBCI-SHS is 96.42%. The system uses federated learning and IoT sensors to predict potential health issues based on real-time analysis of patient data in Eq. 23. This improves early detection and enables proactive interventions, leading to better patient outcomes and enhanced healthcare management, as shown in Fig. 9.

$$f_{xer} [ki - sn''] \rightarrow Ns' - bw [s' + uyr''] + nsad'' . \quad (23)$$

Within the FBCI-SHS framework, the Eq. 23 represents the system's $nsad''$ error recovery procedure (f_{xer}) and fault tolerance ($[ki - sn'']$).

The analysis of proactive healthcare management in the FBCI-SHS shows a 98.37% efficiency in anticipating health risks in Eq. 24. Proactive healthcare management refers to timely predicting and intervening in potential health risks using federated learning outputs. Then, it must be quantified based on measurable indicators such as early anomaly detection rates, patient triage success, or system response time to real-time alerts.

Input Set of IoT devices $\{D_1, D_2, \dots, D_N\}$, initial global model M_0 ,

number of rounds R , learning rate η , epochs E , blockchain ledger B

Output Final global model M^*

- 1: Initialize global model $M \leftarrow M_0$
- 2: for each round $r = 1$ to R do
- 3: for each device $D_i \in \{D_1, D_2, \dots, D_N\}$ in parallel, do
- 4: Retrieve local data D_i
- 5: Train local model $M_i \leftarrow \text{LocalTrain}(M, D_i, \eta, E)$
- 6: Sign model update $\sigma_i \leftarrow \text{Sign}(M_i)$
- 7: Submit (M_i, σ_i) to blockchain B
- 8: end for
- 9: Wait for blockchain validation and consensus
- 10: Retrieve all verified models $\{M_1, M_2, \dots, M_N\}$ from B
- 11: Aggregate global model $M \leftarrow \text{Aggregate}(\{M_1, M_2, \dots, M_N\})$
- 12: end for
- 13: Return the final model M^*

Algorithm 1. Federated Blockchain-IoT framework for sustainable healthcare systems (FBCI-SHS).

Through continuous monitoring of patient's vital signs through IoT devices and ML models, the system detects early signs of potential issues, enabling timely interventions and preventing long-term health complications, thus enhancing overall patient well-being, as shown in Fig. 10.

$$b_f r [x - z n a''] \rightarrow K d [d i - n''] + j S [a f - 9 b w q''] . \quad (24)$$

The system's fault detection and error correction mechanisms inside $[x - z n a'']$ the equation represents the FBCI-SHS architecture $(b_f r)$. In contrast to $K d [d i - n'']$ which simulates the interplay of blockchain verification, $j S [a f - 9 b w q'']$ focuses on efficient data transfer and integrity. Equation 24 exemplifies the framework's commitment to ensuring safe data interchange and robust error handling in the analysis of proactive healthcare management.

The interoperability analysis of FBCI-SHS was exceptionally impressive, as it achieved a score of 96.74%. Due to the ability to incorporate numerous IoT devices, healthcare systems, and data sources, as shown in Eq. 25, the system can facilitate smooth communication and information sharing between diverse systems. This enables the

Metrics	Description	Tools used
IoT Devices	Wearable health-monitoring devices (e.g., smartwatch sensors) collect patient health data.	100 simulated devices (smartwatches, ECG sensors, BP monitors)
Blockchain Framework	Distributed ledger for secure, immutable storage of healthcare data.	Hyperledger Fabric v2.4 with Practical Byzantine Fault Tolerance (PBFT)
Federated Learning	Decentralized ML approach for training models on devices without sharing raw data.	TensorFlow Federated (TFF) with 20 local epochs per round
Simulation Platform	Cloud-based or edge computing platform for running federated learning and blockchain protocols.	Google Cloud Edge Simulator, 8 vCPUs, 32GB RAM, Ubuntu 20.04
Data Types	Anonymized health data (e.g., heart rate, blood pressure, glucose levels, sleep patterns).	Heart rate, blood pressure, glucose levels, ECG signals, sleep cycles
Security Measures	Encryption, secure multi-party computation, and consensus algorithms to ensure privacy and integrity.	AES-256 encryption, Differential Privacy ($\epsilon = 0.3$), Secure Multi-party Computation
Latency	Time delay in real-time response across edge nodes and cloud.	Avg: 315ms per transaction (blockchain), 120ms per FL update

Table 3. The simulation environment.

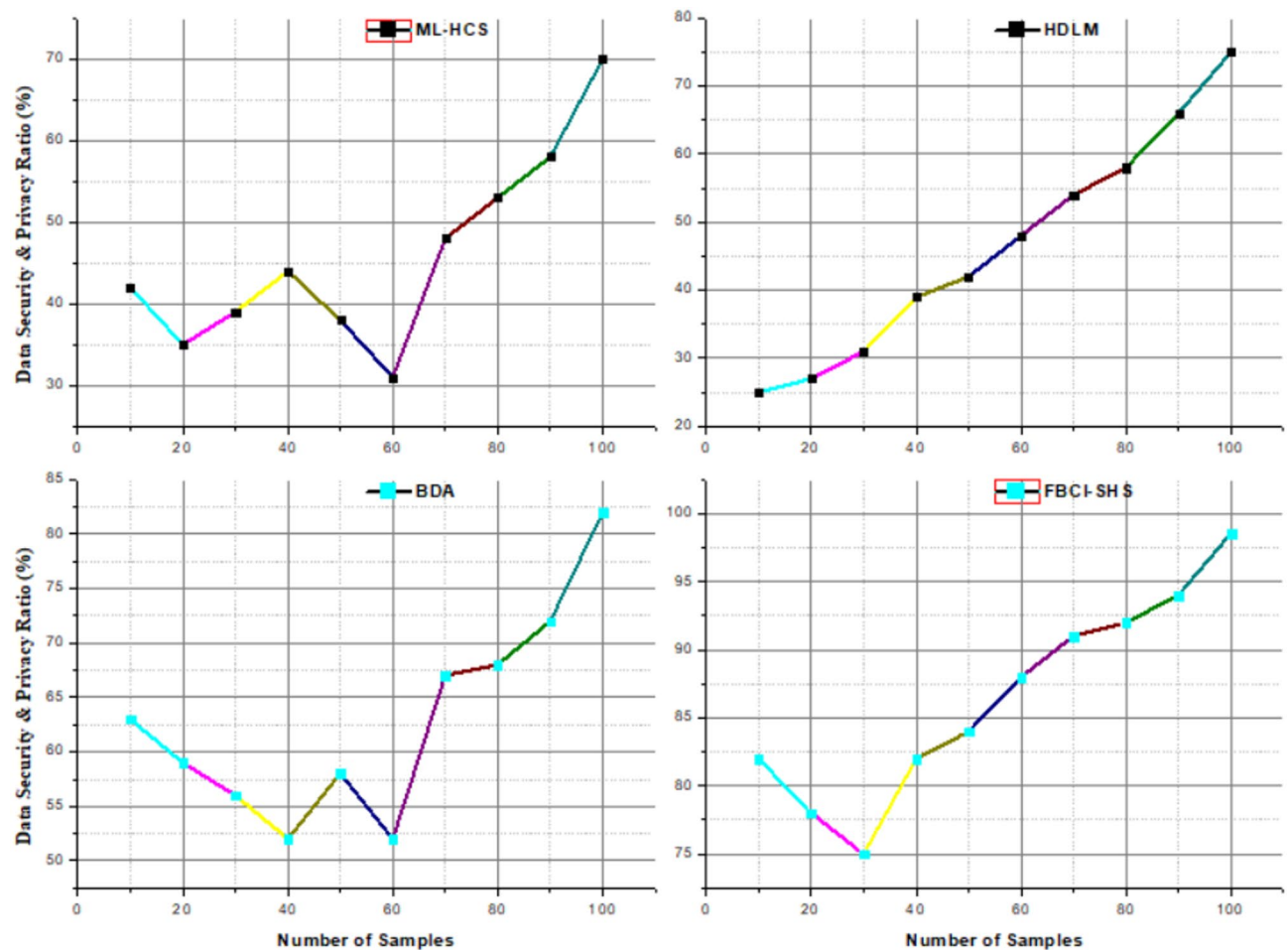


Fig. 7. The analysis of data privacy and security.

efficient collaboration of health care providers with real-time patient monitoring and effective managed care, as shown in Fig. 11.

$$d[mn' - vsq''] \rightarrow Ns[d - 9bd''] + ksi[akw'' + saw''] . \quad (25)$$

The FBCI-SHS framework's management of information is represented by the Eq. 25 ($d[mn' - vsq'']$). $Ns[d - 9bd'']$ represents the integration of sensor data and adaptive learning methods, whereas $ksi[akw'' + saw'']$ ensures confidentiality and integrity via blockchain-based data storage.

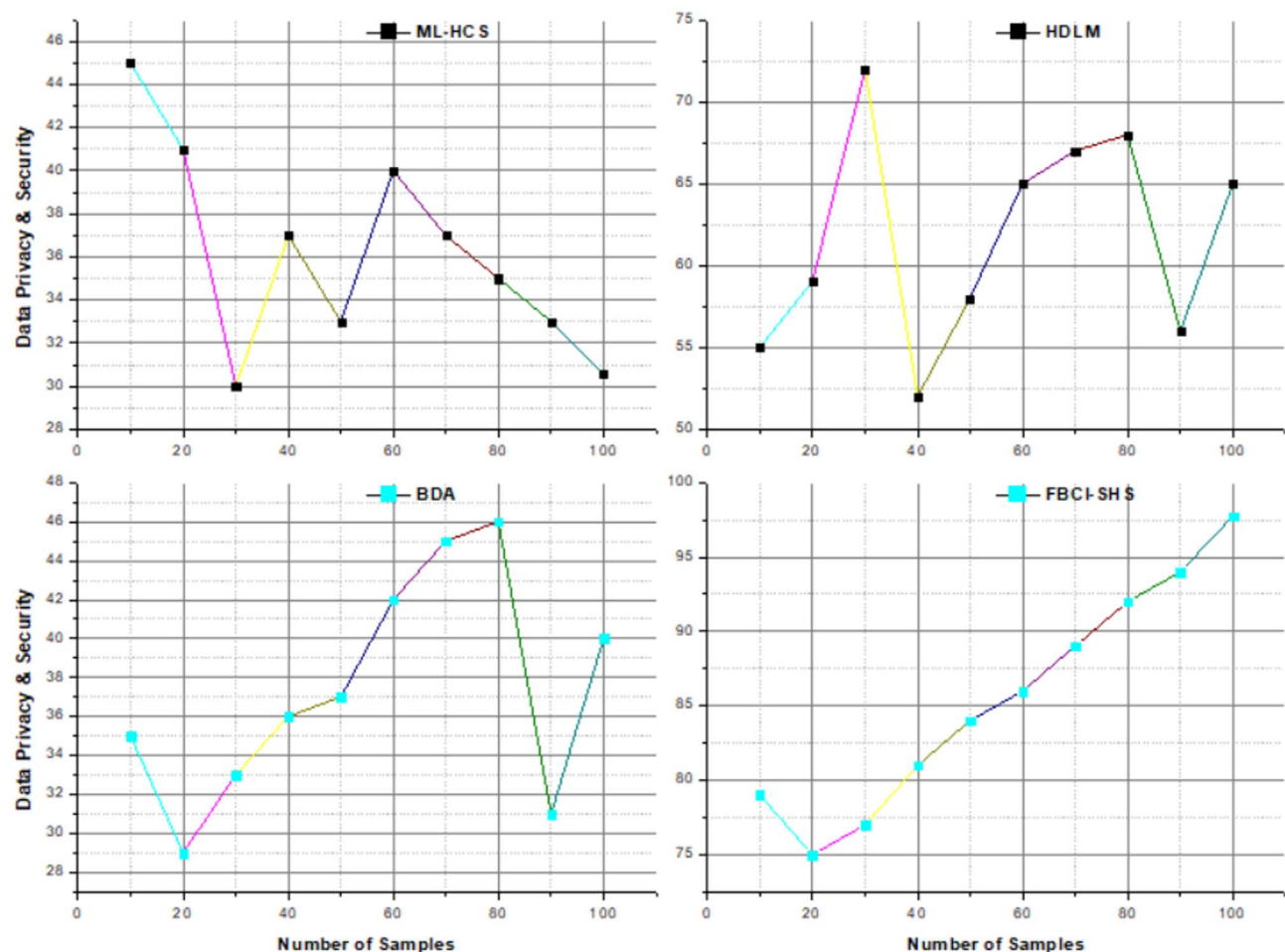


Fig. 8. The analysis of intrusion detection efficiency.

Latency

FBCI-SHS introduces higher initial latency due to the blockchain consensus mechanism (e.g., Practical Byzantine Fault Tolerance or Proof-of-Authority), which verifies and logs model updates. In contrast, PKI-based FL typically exhibits lower latency, as public/private key signing and certificate verification are computationally lighter. However, certificate chain validation in PKI can become a bottleneck in high-volume scenarios or edge deployments. Simulation shows that FBCI-SHS incurs a 25–35% latency increase during model aggregation rounds, while PKI-based FL systems maintain sub-100ms response times in ideal networks.

Resilience to single-point failures

FBCI-SHS offers higher resilience due to the decentralized nature of blockchain. Even if some nodes fail or behave maliciously, the system maintains state consistency via distributed consensus. However, PKI-based FL systems rely on centralized Certificate Authorities (CAs) or identity managers, making them vulnerable to single-point failures such as CA compromise or downtime. Simulated scenarios involving certificate revocation in PKI show immediate disruption across clients until updated CRLs or OCSP responses are fetched, while FBCI-SHS remains functional even with partial network degradation.

Overhead (computational cost)

PKI incurs overhead in digital signing and signature verification at each communication round, with average client-side CPU consumption increasing by 8–15% during high-frequency model updates. FBCI-SHS, while more resource-intensive during consensus rounds, offloads trust establishment from client devices to the blockchain layer, reducing repeated signature processing on edge nodes. However, blockchain consensus (e.g., PBFT) sometimes introduces network-wide communication complexity $O(n^2)$, which becomes expensive as the number of participating clients increases.

Attack resilience and trade-offs

PKI systems under certificate revocation attacks face immediate identity trust breakdown, potentially stalling the entire federated process until new keys/certificates are issued. FBCI-SHS, under 51% attacks (where an adversary

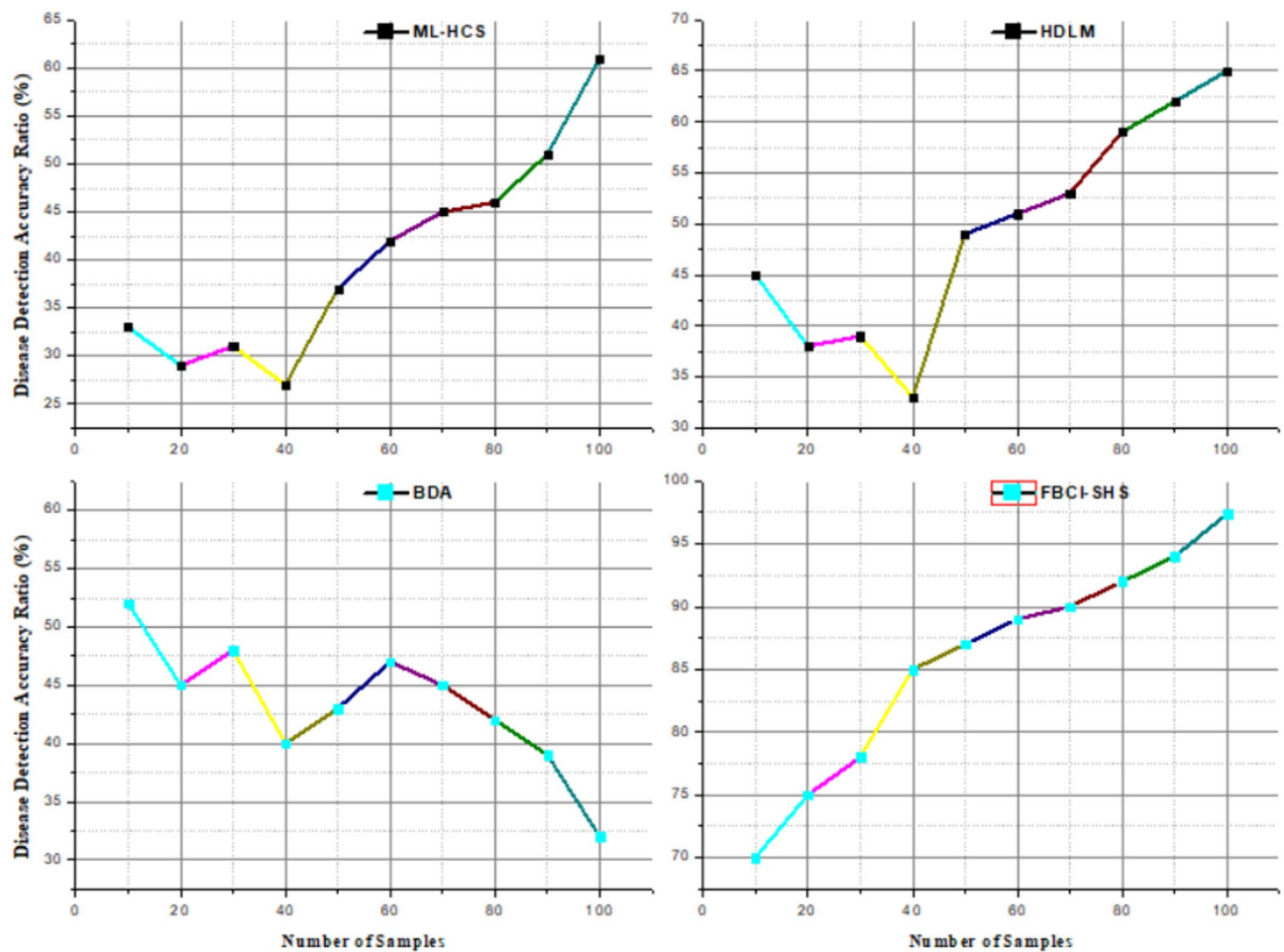


Fig. 9. The analysis of disease detection accuracy.

controls most validator nodes), could suffer from model poisoning or delayed ledger updates. However, such attacks are difficult to execute in permissioned blockchains where validator access is tightly controlled (Table 4).

The FBCI-SHS system incorporates a permissioned ledger, such as Hyperledger Fabric or Quorum, as the blockchain component. This allows for approved and secure involvement from all parties involved in healthcare. Integrity, auditability, and traceability of data are maintained by recording model modifications from federated learning nodes on this decentralized ledger. Key operations, including authenticating signed model changes, implementing access control regulations, and triggering aggregation procedures upon consensus, may be automated using smart contracts. Regarding real-time healthcare settings, a lightweight consensus method like Practical Byzantine Fault Tolerance (PBFT) is ideal since it provides low-latency validation while being energy efficient. The framework's use of zero-knowledge proofs, homomorphic encryption, and differential privacy guarantees that no sensitive patient data will ever be disclosed or kept on-chain, thereby maintaining privacy. Model metadata, signatures, and cryptographic hashes are the only parts of the model saved on the blockchain; off-chain encryption is used for the actual data and model weights. A hybrid on-chain/off-chain storage strategy is instead used. The block structure contains timestamps, anonymized node identities, hashes of model updates, and contract logs for safe tracing.

The FBCI-SHS architecture offers a new perspective on integrating blockchain and federated learning (FL) for the long-term sustainability of healthcare systems, yet it is not without limitations. First, real-time health monitoring may be impeded by the significant computational burden of combining blockchain with FL, particularly at edge devices with limited processing capacity. Furthermore, time-critical healthcare applications may be unable to use blockchain because of its intrinsic delay caused by consensus methods and transaction validation. Another issue is scalability; keeping the network in sync with model changes and maintaining consensus may become resource-intensive as the number of IoT nodes grows.

Conclusion

A FBCI-SHS is an innovative concept that brings federated learning, blockchain, and IoMT to tackle significant security, privacy, and regulatory compliance challenges within healthcare systems. By decentralizing data processing through blockchain-based tamper-proof records, FBCI-SHS ensures confidentiality and promotes trust in healthcare networks. Implementing an IDS would enhance the system's resistance against any possible

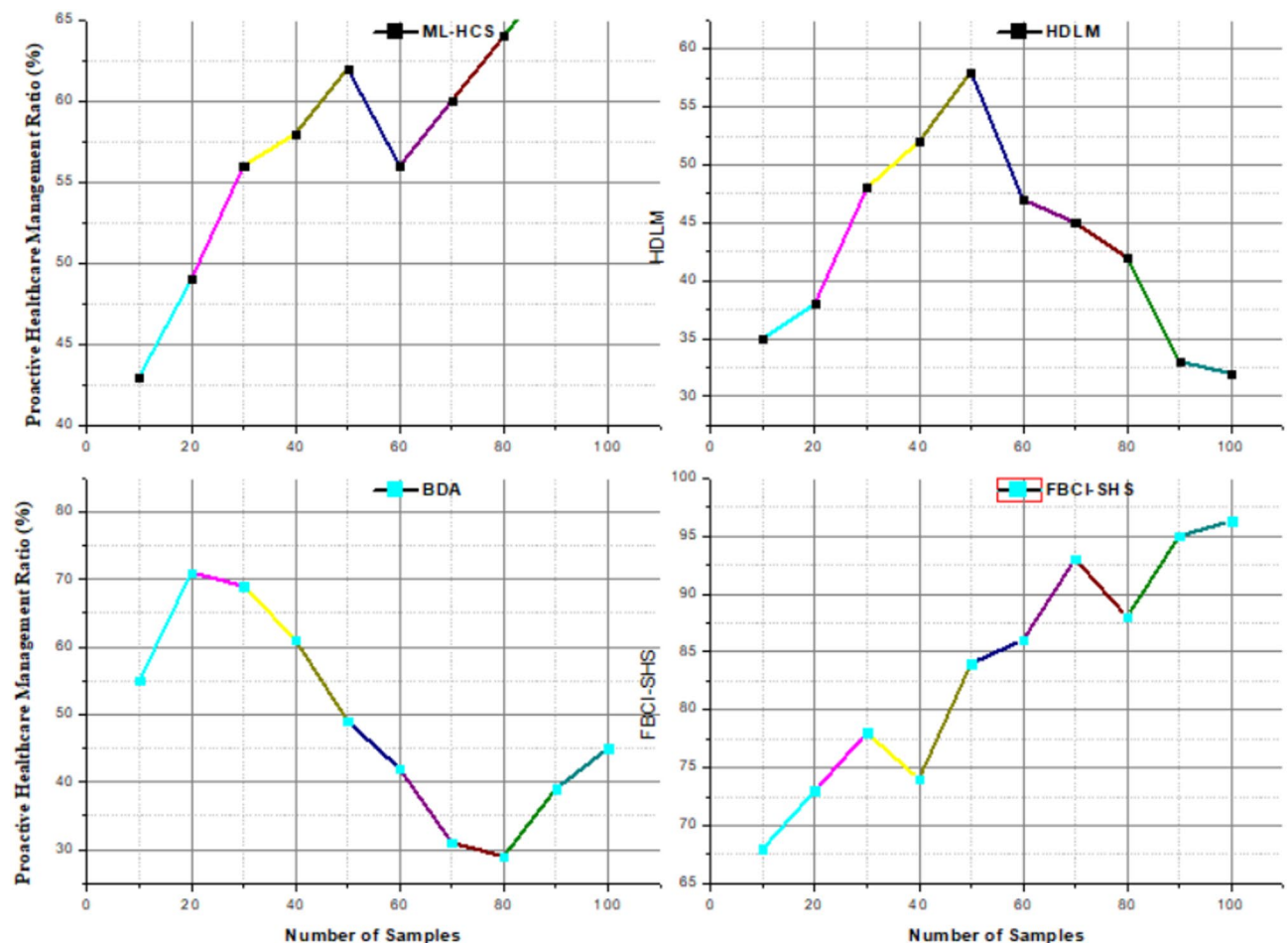


Fig. 10. The analysis of proactive healthcare management.

cyber attack. The suggested framework showed great predictive accuracy for the detection of diseases and intrusion prevention. Therefore, patient care will be optimized, and operation efficiency can be enhanced. This work signifies that AI-driven blockchain-based solutions could build sustainable, secure, and intelligent healthcare systems for Healthcare 5.0. The proposed method achieves data privacy and security by 98.73%, intrusion detection efficiency by 97.16%, disease detection accuracy by 96.425, proactive healthcare management by 98.37%, and interoperability by 96.74% compared to other existing models.

Future work

Involves further development and refinement of the FBCI-SHS framework by integrating cutting-edge ML techniques to obtain more accurate disease predictions and real-time monitoring. This will also investigate real-world deployments across various healthcare settings, assessing its scalability and performance. Further studies will be performed on the full integration of edge computing and look into cross-industry applications of federated learning and blockchain for safe data management.

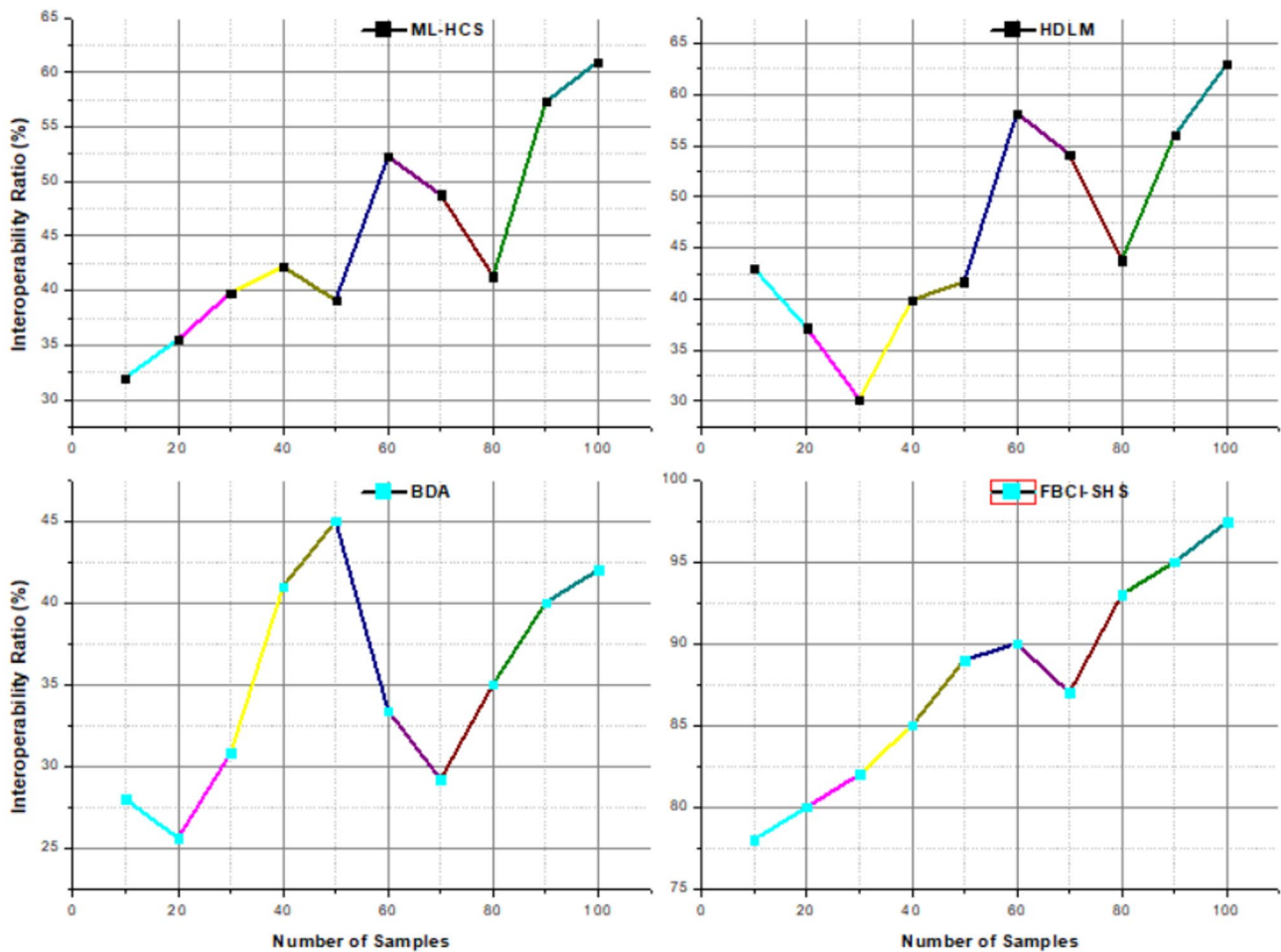


Fig. 11. The analysis of interoperability.

Aspects	Key features	ML-HCS	HDLM	BDA	Proposed method in ratio (%)
Data privacy and security	Utilizes federated learning, blockchain, and encryption to ensure data privacy	67.2%	78.24%	33.53%	98.73%
High intrusion detection efficiency	Employs advanced ML models and real-time monitoring, achieving a detection rate	87.5%	78.54%	36.82%	97.16%
Accurate disease detection	Leverages federated learning and IoT sensors, achieving accuracy in predicting potential	57.2%	48.44%	39.05%	96.42%
Proactive healthcare management	Continuously monitors vital signs, enabling efficiency in early risk anticipation	37.2%	88.35%	41.26%	98.37%
Seamless interoperability	Ensures smooth communication across various IoT devices, healthcare platforms, and data sources	57.3%	65.72%	43.67%	96.74%

Table 4. The comparison of exiting methods and proposed method.

Data availability

The datasets generated and/or analyzed during the current study are available from Kaggle: <https://www.kaggle.com/datasets/faismalik/iot-healthcare-security-dataset>.

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Conceptualization, B.B, P.M.R.; Writing—Review and Editing, P.S, S.G.K.P; Methodology, J.K.B, S.I; Supervision, M.K., A.F.E.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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